

C1002 — Programming Fundamentals | PF Lab Tasks

FAST — National University of Computer and Emerging Sciences Karachi campus

Department of Software Engineering

PF LAB TASKS

Course Code: C1002

Subject: Programming Fundamentals

Section: C

Instructor: Mr. Shayan Ali

STUDENT SUBMISSION DETAILS

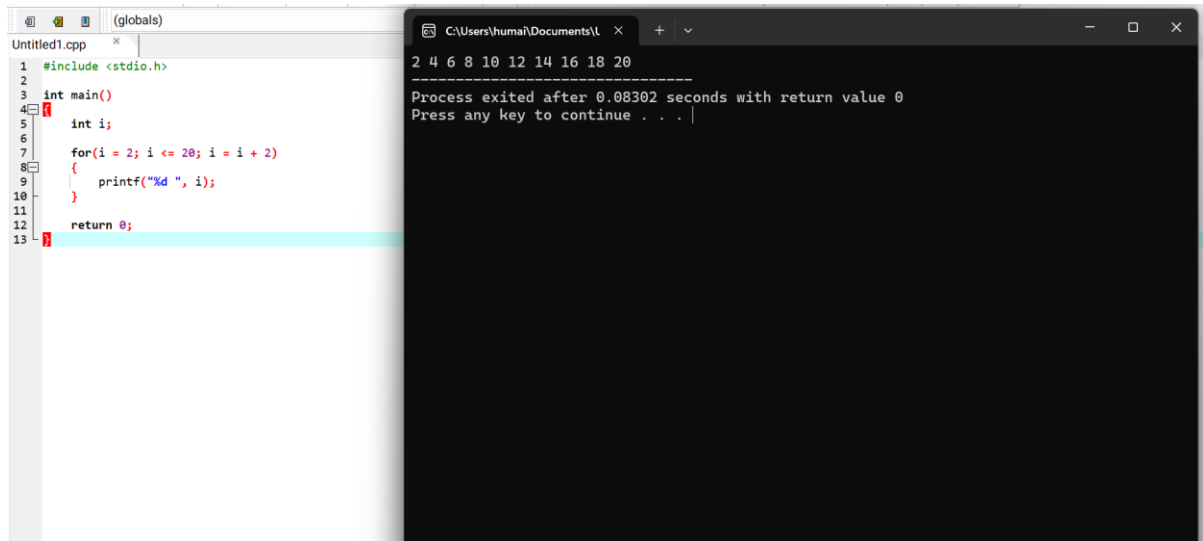
Name: AMNA KHAN

Roll No / ID: 26K-3091

Class / Section: C

Date of Submission: 30-SEPTEMBER'2026

Task 1

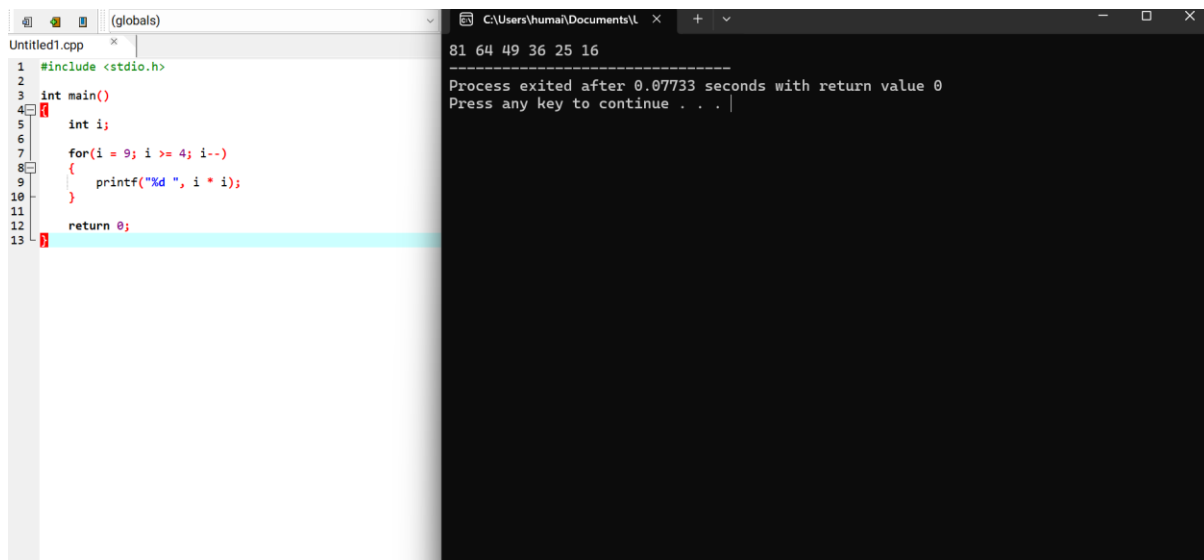


The image shows a side-by-side comparison of a C++ source code file and its execution output. On the left, a code editor window titled 'Untitled1.cpp' displays the following code:

```
1 #include <stdio.h>
2
3 int main()
4 {
5     int i;
6
7     for(i = 2; i <= 20; i = i + 2)
8     {
9         printf("%d ", i);
10    }
11
12    return 0;
13 }
```

On the right, a terminal window shows the output of the program. The first line displays the sequence of numbers: '2 4 6 8 10 12 14 16 18 20'. The second line shows the program's exit status: 'Process exited after 0.08302 seconds with return value 0'. The third line prompts the user to continue: 'Press any key to continue . . . |'.

Task 2



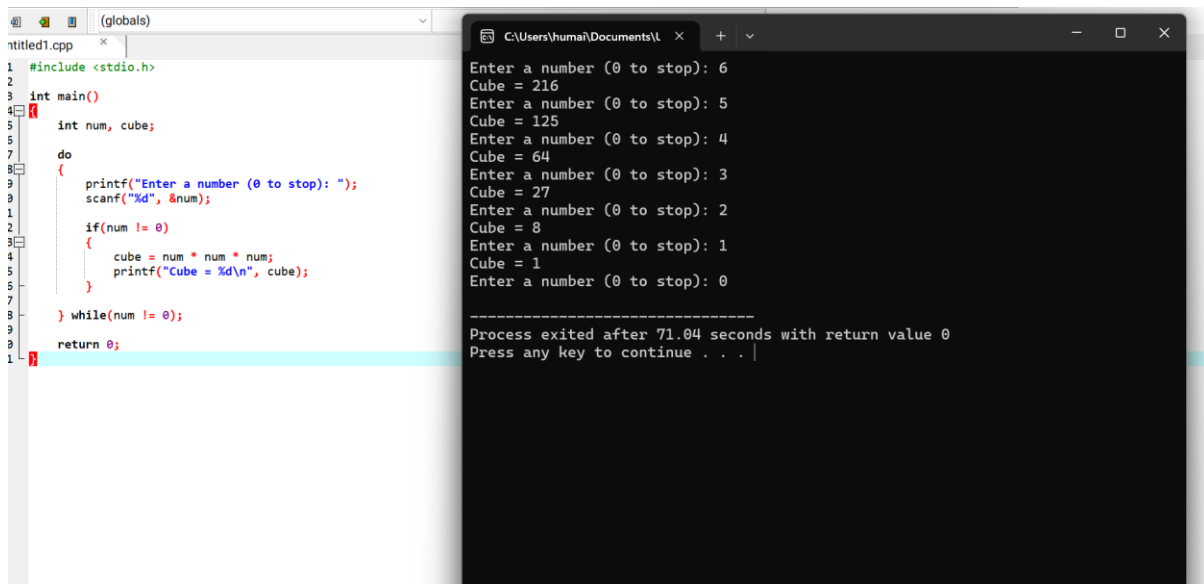
The screenshot shows a C++ IDE with two windows. The left window, titled 'Untitled1.cpp', contains the following code:

```
1 #include <stdio.h>
2
3 int main()
4 {
5     int i;
6
7     for(i = 9; i >= 4; i--)
8     {
9         printf("%d ", i * i);
10    }
11
12    return 0;
13 }
```

The right window shows the output of the program:

```
81 64 49 36 25 16
-----
Process exited after 0.07733 seconds with return value 0
Press any key to continue . . .
```

Task 3



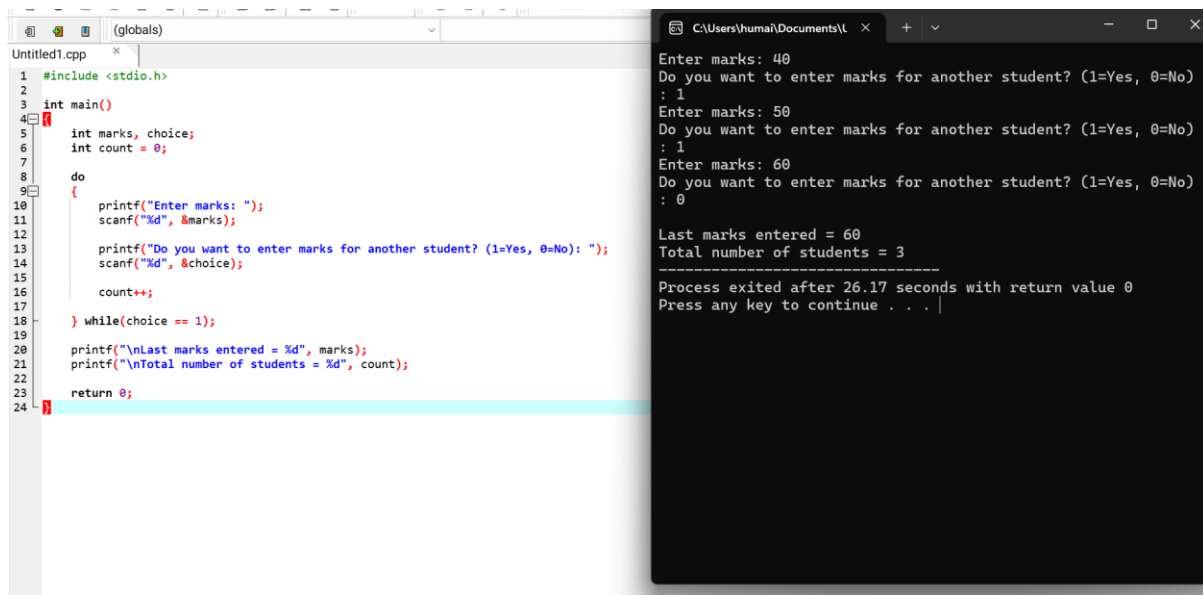
The screenshot shows a C++ IDE with two windows. The left window, titled 'Untitled1.cpp', contains the following code:

```
1 #include <stdio.h>
2
3 int main()
4 {
5     int num, cube;
6
7     do
8     {
9         printf("Enter a number (0 to stop): ");
10        scanf("%d", &num);
11
12        if(num != 0)
13        {
14            cube = num * num * num;
15            printf("Cube = %d\n", cube);
16        }
17    } while(num != 0);
18
19    return 0;
20 }
```

The right window shows the output of the program:

```
Enter a number (0 to stop): 6
Cube = 216
Enter a number (0 to stop): 5
Cube = 125
Enter a number (0 to stop): 4
Cube = 64
Enter a number (0 to stop): 3
Cube = 27
Enter a number (0 to stop): 2
Cube = 8
Enter a number (0 to stop): 1
Cube = 1
Enter a number (0 to stop): 0
-----
Process exited after 71.04 seconds with return value 0
Press any key to continue . . .
```

Task 4

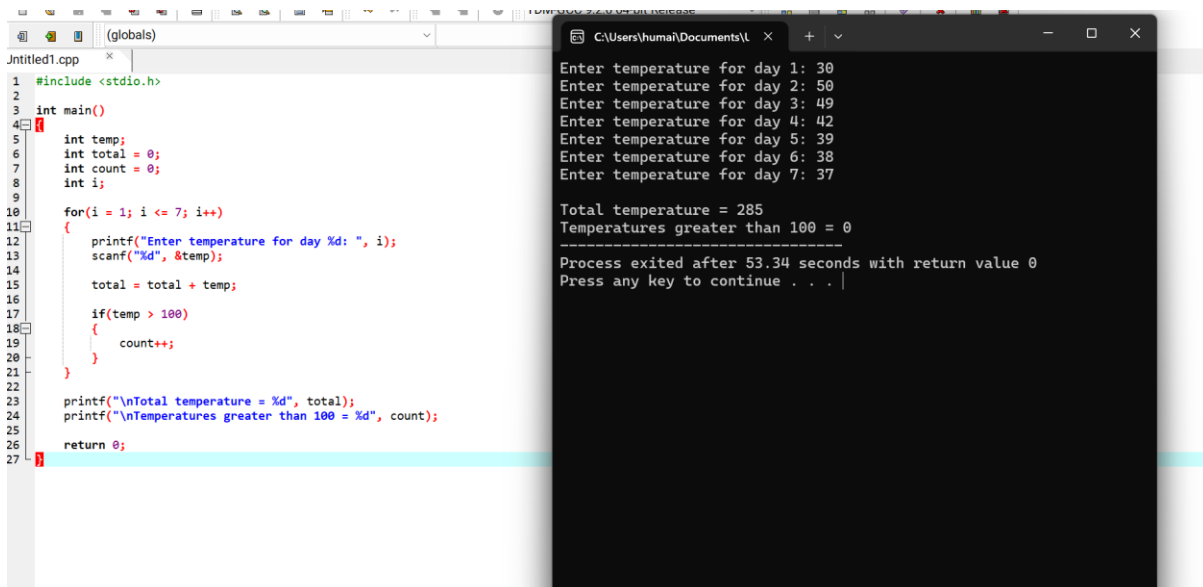


```
1 #include <stdio.h>
2
3 int main()
4 {
5     int marks, choice;
6     int count = 0;
7
8     do
9     {
10         printf("Enter marks: ");
11         scanf("%d", &marks);
12
13         printf("Do you want to enter marks for another student? (1=Yes, 0=No): ");
14         scanf("%d", &choice);
15
16         count++;
17     } while(choice == 1);
18
19     printf("\nLast marks entered = %d", marks);
20     printf("\nTotal number of students = %d", count);
21
22     return 0;
23 }
```

```
Enter marks: 40
Do you want to enter marks for another student? (1=Yes, 0=No)
: 1
Enter marks: 50
Do you want to enter marks for another student? (1=Yes, 0=No)
: 1
Enter marks: 60
Do you want to enter marks for another student? (1=Yes, 0=No)
: 0

Last marks entered = 60
Total number of students = 3
-----
Process exited after 26.17 seconds with return value 0
Press any key to continue . . . |
```

Task 5



```
1 #include <stdio.h>
2
3 int main()
4 {
5     int temp;
6     int total = 0;
7     int count = 0;
8     int i;
9
10    for(i = 1; i <= 7; i++)
11    {
12        printf("Enter temperature for day %d: ", i);
13        scanf("%d", &temp);
14
15        total = total + temp;
16
17        if(temp > 100)
18        {
19            count++;
20        }
21    }
22
23    printf("\nTotal temperature = %d", total);
24    printf("\nTemperatures greater than 100 = %d", count);
25
26    return 0;
27 }
```

```
Enter temperature for day 1: 30
Enter temperature for day 2: 50
Enter temperature for day 3: 49
Enter temperature for day 4: 42
Enter temperature for day 5: 39
Enter temperature for day 6: 38
Enter temperature for day 7: 37

Total temperature = 285
Temperatures greater than 100 = 0
-----
Process exited after 53.34 seconds with return value 0
Press any key to continue . . . |
```

Task 6

The screenshot shows a C++ IDE with a file named 'Untitled1.cpp'. The code is as follows:

```
1 #include <stdio.h>
2
3 int main()
4 {
5     int amount;
6     int total = 0;
7     int deposits = 0;
8
9     while(1)
10    {
11        printf("Enter amount saved: ");
12        scanf("%d", &amount);
13
14        if(amount <= 0)
15        {
16            break;
17        }
18
19        total = total + amount;
20        deposits++;
21    }
22
23    printf("\nTotal savings = %d", total);
24    printf("\nNumber of deposits = %d", deposits);
25
26    return 0;
27 }
```

The output window on the right shows the following text:

```
Enter amount saved: 500
Enter amount saved: 600
Enter amount saved: 0

Total savings = 1100
Number of deposits = 2
-----
Process exited after 21.02 seconds with return value 0
Press any key to continue . . . |
```

Task 7

The screenshot shows a C++ IDE with a file named 'Untitled1.cpp'. The code is as follows:

```
1 #include <stdio.h>
2
3 int main()
4 {
5     char name[50];
6     float price, total = 0;
7     int choice;
8     int items = 0;
9
10    do
11    {
12        printf("\nEnter food name: ");
13        scanf("%49[^\n]", name);
14
15        printf("Enter price: ");
16        scanf("%f", &price);
17
18        total = total + price;
19        items++;
20
21        printf("Do you want to order another item? (1=Yes, 0=No): ");
22        scanf("%d", &choice);
23    } while(choice == 1);
24
25    printf("\nTotal bill = %.2f", total);
26    printf("\nNumber of items ordered = %d", items);
27
28    return 0;
29 }
```

The output window on the right shows the following text:

```
Enter food name: biryani
Enter price: 250
Do you want to order another item? (1=Yes, 0=No): 0

Total bill = 250.00
Number of items ordered = 1
-----
Process exited after 44.36 seconds with return value 0
Press any key to continue . . . |
```

Task 8

```
Untitled1.cpp
1 #include <stdio.h>
2
3 int main()
4 {
5     int salary[6];
6     int i;
7     int count = 0;
8
9     for(i = 0; i < 6; i++)
10    {
11        printf("Enter salary of employee %d: ", i + 1);
12        scanf("%d", &salary[i]);
13
14        if(salary[i] > 50000)
15        {
16            count++;
17        }
18    }
19
20    printf("\nEmployee salaries:\n");
21
22    for(i = 0; i < 6; i++)
23    {
24        printf("Employee %d = %d\n", i + 1, salary[i]);
25    }
26
27    printf("\nNumber of employees with salary greater than 50000 = %d", count);
28
29    return 0;
30 }
```

```
C:\Users\humah\Documents\L x + -
Enter salary of employee 1: 100000
Enter salary of employee 2: 30000
Enter salary of employee 3: 50000
Enter salary of employee 4: 89000
Enter salary of employee 5: 90000
Enter salary of employee 6: 65000

Employee salaries:
Employee 1 = 100000
Employee 2 = 30000
Employee 3 = 50000
Employee 4 = 89000
Employee 5 = 90000
Employee 6 = 65000

Number of employees with salary greater than 50000 = 4
-----
Process exited after 64.54 seconds with return value 0
Press any key to continue . . . |
```